

Basel Ahmed Sardah

AI Engineer and Computer Vision Specialist focusing on deep learning architectures, object detection, and real-time GPU inference. Proven expertise in engineering end-to-end vision pipelines (from dataset augmentation to TensorRT deployment), building containerized API services, and executing industrial anomaly detection solutions to drive production-grade AI stability.

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Technical Projects

QASSAS (Industrial Visual Anomaly Detection) | PyTorch, Diffusion Models, OpenCV, NumPy 2025 - 2026

- Engineered an end-to-end Computer Vision pipeline for micro-defect detection in industrial manufacturing, achieving an 87% overall accuracy and a 97% Recall rate.
- Prepared and managed high-dimensional datasets, formulating rigorous data preprocessing, cleaning, and augmentation logic using NumPy to maintain maximum dataset integrity during model training.
- Executed post-processing optimization and integration testing, ensuring the stability and reproducibility of the vision-based AI system for production environments.

Edge AI Vision System (Real-Time Inference) | Python, YOLO, TensorRT, Linux, OpenCV 2026

- Architected an asynchronous edge AI pipeline using NVIDIA TensorRT (FP16), decoupling I/O frame acquisition and deep learning inference to sustain 60+ FPS video analytics without CPU starvation.
- Engineered a deterministic spatial decision engine for hardware actuation, applying Euclidean hazard avoidance algorithms and object tracking for automated trajectory mapping.
- Optimized model evaluation and GPU execution speed, transitioning raw PyTorch models into precision-calibrated engines for low-latency, real-world deployment.

AI Trust Gateway (NLP & Feature Extraction) | Python, PyTorch, XGBoost 2026

- Engineered a fault-tolerant deep learning pipeline using a Sliding Window Chunking mechanism to bypass Out-Of-Memory (OOM) bottlenecks during high-volume inference.
- Deployed a low-latency hybrid classifier combining a frozen encoder with XGBoost, drastically reducing compute overhead while achieving a 99.64% binary detection accuracy.

WORK EXPERIENCE

Beaee Platform

Software Engineer Intern

Jan/2026 - Jul/2026

- Deployed API services and backend architectures into production, containerizing server environments utilizing Docker to ensure cross-platform reproducibility and pipeline consistency.
- Collaborated closely with cross-functional teams to align technical data solutions with organizational goals, resolving manual administrative bottlenecks through Python-based workflow automation.
- Led database re-architecture initiatives, optimizing data normalization to accelerate querying performance for robust analytical reporting.

KFU Clubs | Technical Instructor (Algorithms & Math for CS)

2023 – 2026

- Delivered over 30 hours of instructional workshops on Algorithm Analysis, clearly communicating complex computational and engineering concepts to collaborative tech teams.

Technical Skills

- Computer Vision & AI: PyTorch, TensorFlow, OpenCV, YOLO, NVIDIA TensorRT, CNNs, Deep Learning, Object Detection, Image Segmentation.
- Data Engineering & MLOps: Pandas, NumPy, Docker, Containerized APIs, Dataset Labeling & Augmentation, Git/GitHub.
- Programming & Infrastructure: Python, SQL, FastAPI, Linux (Arch/Wayland), GPU-based Inference.

EDUCATION and Certifications

- King Faisal University Bachelor of Computer Science with GPA 4.79 / 5.00 (2021 - 2026) .
- AI Machine Learning Specialization from DeepLearningAI (2025).
- Google Data Analytics Professional Certificate (2024).
- University of Michigan Intro to Data Science in Python (2025).